

Markov chain models for forest successions in the Erzgebirge, Germany

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Abstract

Markov chain models have been developed to describe hypothetical and observed forest successions in the Erzgebirge (Ore Mountains), Germany. The first model is time-homogeneous and it is applied to the whole Erzgebirge area (altitude up to 900 m). The natural course of succession in this area is hypothetical since forest management over the last 200 years has been depriving the forest vegetation of any chance to develop in its natural way. The second model is time-inhomogeneous and it is applied to an area where secondary succession has really begun after the devastation of spruce stand due to strong SO₂-pollution (especially in the 1980s) and removal of spruce from the forest. Taking possible human impact into account, the model can predict vegetation development in the long-term under particular scenarios of forest management and other impacts. Both examples demonstrate that Markov chain models represent an efficient tool to organise the existing knowledge of forest succession into a system of quantitative predictions.

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1. Introduction

The Erzgebirge (Ore Mountains) is a mountain area in Saxony, Germany located along the border to Czech Republic (Fig. 1). Some parts of the area were strongly damaged by SO₂ pollution in the 1980s. Although forest successions are under

careful study here (Schmidt, 1993, 1995; Schmidt et al., 1997, 2001; Thomasius, 1981, 1990, 1996), Markov models have not been applied in Saxony up to now. This tool has only been used in some case studies of grassland vegetation in Germany (Balzer, 1998, 2000). The aim of this paper is to find out in principle in what respect the application of Markov models may support management decision making and understanding of long-term development of forest areas in the Erzgebirge, especially for damaged spots.

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Fig. 1. Geographic location of study areas in the Erzgebirge (Ore Mountains): Altenberg, Olbernhau.

Studies of regularities in vegetation dynamics have a long tradition. They originate from a theory laid down by Clements (1916, 1936), who formulated a concept of dynamic vegetation system as a structure ordered in time. Later, from comparative studies of the Swedish, Moscow, French–Swiss and English–American schools of geobotany, Razumovsky (1981) developed an idea of strict determination in the temporal changes of vegetation and a hypothesis on general regularities of those changes within a botanic–geographic region.

Succession means that the vegetation of an area develops through a number of different plant communities (stages) in the direction to the climax (final) stage which is in balance with the climate of the region and other relevant conditions. The process of climax acquisition takes a long time and is associated with drastic changes in the species composition. The succession appears if changes are caused by the vital activity of the plant community and realised through the mechanisms of interspecies competition. The inter-

mediate (seral) stages of succession depend on various ecological conditions such as humidity and soil richness (Schmidt, 1993, 1995).

Successions can be affected by both anthropogenic and natural impacts. In any example of diverse anthropogenic impacts, one can observe a fairly restricted set of communities through which the vegetation develops in response to the impact. As a result, there is a finite number of possible succession stages present in a particular area, and the climax state may even not be reachable (Korotkov et al., 2001). When the reason for the climax non-acquisition is in the human pressure, the terminal state of succession is called a sub-climax, or anthropogenic climax after Tansley (1935).

The similarity between transitions in successional processes directed from different pioneer stages to defined climax stages, on one hand, and transitions in Markov-chain processes directed from different initial states to defined final states, on the other hand, stimulates one to use Markov-chain models for the simulation of succession

processes (Logofet, 1999). Markov chain models of forest successions became such a popular paradigm since late 1970s (Horn, 1975, 1981; Usher, 1979, 1981).

2. Succession scheme for the Erzgebirge vegetation

It is hypothetically supposed in this section that there no anthropogenic impact on the forest area. Climatic conditions of the Erzgebirge area (Fig. 1) depend greatly on the altitude. They are reflected in the composition of natural forest communities. Rowan–spruce forests (*Sorbo–Piceetum*) dominate at moist and cool locations of the mountain ridges. As the altitude decreases and the vegetation period extends, dominant positions on the sites devoid of ground and stagnant water are handed over from spruce forest to montane beech–spruce forest (*Fago–Piceetum*). Substrate-demanding communities of *Fago–Piceetum* which had been already interspersed with fir have been partly growing up with sycamore nowadays. In *Fagetum* communities on medium elevations, beech performs a more important role, while the proportion of spruce continues to decline. In lowlands, spruce completely retreats from natural forest communities, and the dominant role is assumed by beech. Together with beech, the following species occur: ash (*Fraxinus excelsior*), sycamore (*Acer pseudo-platanus*) and seldom elm (*Ulmus glabra*) on moist and cool sites, e.g. along furrows; sessile oak (*Quercus petraea*) on the sites with dry and warm climate, viz. on sun-exposed slopes and arching terrains; large-leaved lime (*Tilia platyphyllos*) on rich soils; hornbeam (*Carpinus betulus*) and wild cherry (*Prunus avium*). The upland form of Scots pine (*Pinus sylvestris*) is represented in the natural forest associations due to the bedrock substrate which is poor and greatly permeable (high proportion of coarse sand and skeletal soil). It occurs on dome-shaped and sun-exposed terrains overlying tourmaline granite, quartzitic schist, quartz phyllite, quartz porphyry and lime sandstone. Silviculture has relied heavily on coniferous species over

centuries, in particular, on pure spruce stands. It has impaired not only the natural associations in herb and tree layers, but also the original type of humus (Hunger, 1994).

As a precondition for the construction of Markov model one needs geobotanical information about the potential successional stages and transitional probabilities between them. This information may be visually presented in a successional scheme (see Figs. 2, 4 and 5) which supports effectively interdisciplinary cooperation. The scheme of Fig. 2 has been composed by means of literature (Pott, 1992; Schmidt, 1993, 1995; Schmidt et al., 1997, 2001) and personal communication (P.A. Schmidt). The latter uses a tree ontogeny concept proposed by Evstigneev (Evstigneev et al., 2001; Evstigneev and Pomaz, 2001; Fig. 2).

The potential pathways for successions in the Erzgebirge lead to 6 climax states (nos. 1–6).

3. Homogeneous model for forest successions in the whole Erzgebirge area

Under the assumption that environmental conditions do not change essentially in the Erzgebirge area, a time-homogeneous Markov chain model has been constructed. To construct such a model the transition matrix P and the initial area-distribution vector x are necessary (Kemeny and Snell, 1983). The latter has the dimension of all possible states (30 in our example) and each of its components equal to the relative area of the corresponding successional state.

An element p_{ij} of the transition matrix $P = [p_{ij}]$ denotes the probability of transition from succession state i into j in one time step Δt ($\Delta t = 5$ years in the case under study). The continuous process of successional changes is approximated by transitions between thirty discrete states shown in Fig. 2. Six of them (nos. 1–6) are absorbing states, or climax ones, and 24 states (nos. 7–30) are transient, or successional.

The diagonal matrix elements p_{ii} represent the probability to remain in the same state after one time step Δt . According to Logofet (Knyazkov et al., 1992; Logofet and Lesnaya, 2000), their values

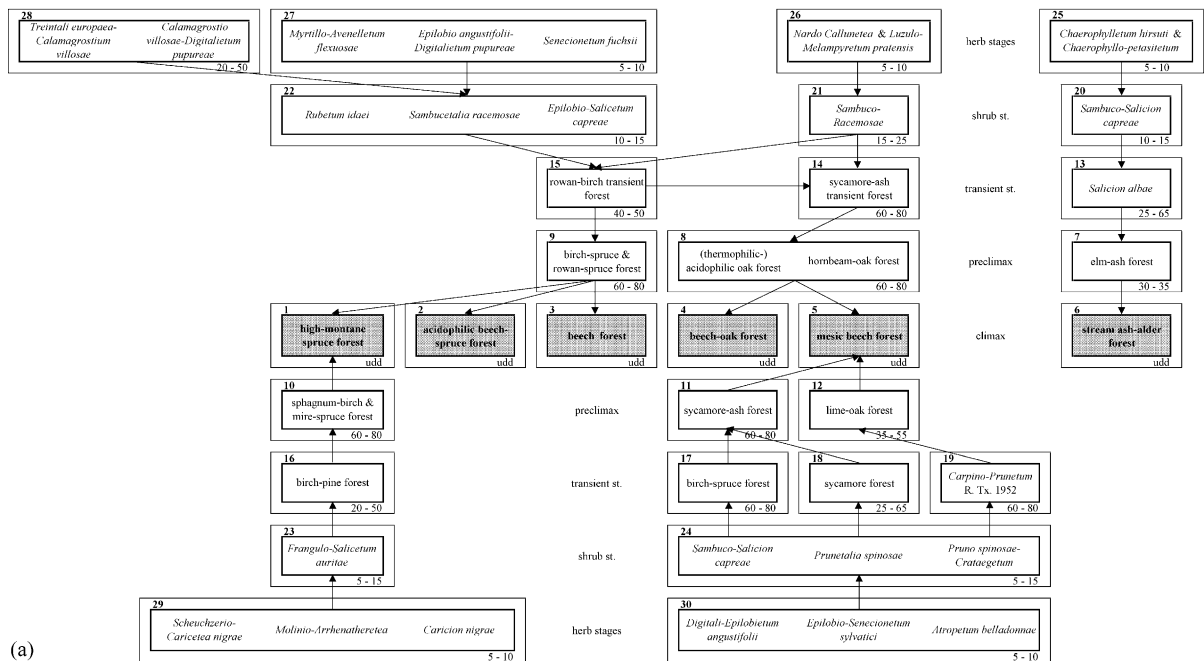


Fig. 2. Construction of the scheme of hypothetical vegetation development (without human impact) for the Erzgebirge. (a) The scheme of vegetation development in the Erzgebirge built with the help of (b–d). States 1–6 last until a drastic disturbance. (b) Geobotanic characteristics of states presented in the scheme of vegetation dynamics in the Erzgebirge built with the help of (c). (c) Ontogenetic periods of dominant species in the Erzgebirge, years (after Brzeziecki and Kienast, 1994; Smirnova, 1994). (d) Species-specific ontogenetic development of communities in the Erzgebirge.

can be obtained if it is known how long, in the average number of time steps m_i , the chain remains in the same state:

$$p_{ii} = 1 - \frac{1}{m_i}, \quad i = 1, \dots, 30 \quad (1)$$

Vector $\mathbf{m}$ (with 24 transient elements), where $\mathbf{m}_{\min}$ and $\mathbf{m}_{\max}$ denote minimal and maximal state-duration times, can be obtained as follows.

Continuous process of individual plant development could be subdivided into several stages by means of morphological indicators which reflect functional importance. Ontogeny of plant organism is subdivided into four periods and nine stages (Smirnova et al., 1999):

- 1) latent period (seed);
- 2) pre-reproductive period (seedling, juvenile, immature, virginal);
- 3) reproductive period (young, mature and old reproductive);

- 4) post-reproductive period (senile),

while two periods and five stages, namely prereproductive period with *immature* (*im*) and *virginal* (*v*) stages and reproductive period with young (*g1*), mature (*g2*) and old (*g3*) generative stages, are significant in our case study.

Development of a plant community can be treated from the standpoint of its individual species ontogeny (Evstigneev et al., 2001; Evstigneev and Pomaz, 2001; Fig. 2). The durations of ontogenetic intervals for Central-European forests and for the Erzgebirge, in particular, are rare in the literature (Harper and White, 1974; Brzezicki and Kienast, 1994; Harper, 1994). As an approximation, the species-specific values for slight-disturbed East-European broad-leaved forests under normal and oppressed tree-stand conditions (Fig. 2) are applied in the case study. These data are obtained through ontogenetic analysis in

No.	Associations (Latin, English) and dominant species (in parentheses)	stage duration intervals, yrs.
1	<i>Calamagrostio villosae-Piceetum</i> (Tx. 37) Hartm. ex Schlut. 66 high-montane spruce forest (Pab dom.)	udd (until a drastic disturbance) indicates that this stage could be transferred to another one by special event (e.g. snow catastrophe or fire) which is not taken into account by the model.
2	<i>Calamagrostio villosae-Fagetum</i> Mikyska 72, <i>Luzulo-Fagion</i> acidophilic beech-spruce forest (Fsy, Pab polydom.; Aps)	
3	<i>Luzulo-Fagetum</i> Meusel 37 beech forest (Fsy dom., Cbe, Aps; Aal in the past)	
4	<i>Luzulo-Fagetum</i> Meusel 37 beech-oak forest (Fsy dom., Qpe, Qro, Cbe, Aps; Aal in the past)	
5	<i>Galio odorati-Fagetum impatientetosum</i> mesic beech forest (Fsy dom.; Aps, Fex, Pab; Aal in the past)	
6	<i>Alno-Fraxinetum</i> Ruhl 64 stream ash-alder forest (Agl, Fex, Sfr, Ugl, Ula)	
7	elm-ash forest	$30 < (g\ 1-im Ugl, n) < 35^*)$
8	<i>Galio-Carpinetum</i> Oberd. 57 (thermophilic-)acidophilic oak forest & hornbeam-oak forest (Fsy, Qpe, Qro polydom.; Tco, Cbe, Aps)	$60 < (g\ 1-im Fsy, n) < 80$
9	<i>Sorbo-Piceetum</i> (birch-spruce & rowan-spruce forest)	$60 < (g\ 1-im Pab, n) < 80$
10	<i>Vaccinio uliginosi-Betuletum pubescentis</i> Libb. 33 sphagnum-birch & mire-spruce forest (Bpu, Pmu)	
11	sycamore-ash forest (Aps, Fex)	$60 < (g\ 1-im Fsy, n) < 80$
12	lime-oak forest	$35 < (g\ 1-im Tco, n) < 55$
13	<i>Salicion albae</i>	$25 < (g\ 1-im Fex, n) < 65$
14	sycamore-ash transient forest	$60 < (g\ 1-im Qro, n) < 80$
15	rowan-birch transient forest	$40 < (g\ 1-im Aps, n) < 50$
16	birch-pine forest	$20 < (g\ 1-im Pmu, n) < 50$
17	birch-spruce forest	$60 < (g\ 1-im Pab, n) < 80$
18	sycamore forest	$25 < (g\ 1-im Fex, n) < 65$
19	<i>Carpino-Prunetum</i> R. Tx. 1952 (Qro, Cbe)	$60 < (g\ 1-im Qro, n) < 80$
20	<i>Sambuco-Salicion capreae</i>	$10 < (v-im Sfr, n) < 15$
21	<i>Sambuco-Racemosae</i>	$15 < (v-im Cbe, n) < 25$
22	<i>Rubetum idaei</i> <i>Sambucetalia racemosae</i> <i>Epilobio-Salicetum capreae</i>	$10 < (v-im Sau, n) < 15$
23	<i>Frangulo-Salicetum auritae</i> <i>Sambuco-Salicion capreae</i>	$5 < (v-im Bpu, n) < 15$
24	<i>Prunetalia spinosae</i> <i>Pruno spinosae-Crataegetum</i>	$5 < (v-im Bpe, n) < 15$
25	<i>Chaerophylletum hirsuti</i> <i>Chaerophyllo-petasitetum</i>	5 - 10
26	<i>Nardo Callunetea</i> <i>Luzulo-Melampyretum pratensis</i> <i>Senecionetum fuchsii</i>	
27	<i>Epilobio angustifolii-Digitalietum pupureae</i> <i>Myrtillo-Avenelletum flexuosae</i>	
28	<i>Calamagrostio villosae-Digitalietum pupureae</i> <i>Treintali europaea-Calamagrostium villosae</i>	5 - 20
29	<i>Scheuchzerio-Caricetea nigrae</i> <i>Molinio-Arrhenatheretea</i> <i>Caricion nigrae</i>	5 - 10
30	<i>Digitali-Epilobietum angustifolii</i> <i>Epilobio-Senecionetum sylvatici</i> <i>Atropetum belladonnae</i>	

*) $30 = 50 - 20 = g\ 1(Ugl, n) - im(Ugl, n)$ and $g\ 1(Ugl, n) - im(Ugl, n) = 36 - 1 = 35$: difference between the young generative and *im* mature ages of *Ulmus glabra* under *n* ormal conditions (see Fig. 2c).

(b)

Fig. 2 (Continued)

different parts of the area (Smirnova, 1994, p. 100). The elements of vectors m_{min} and m_{max} denote durations of successional states (in units

of Δt) calculated by minimal and maximal values of ontogenetic periods. In our case (Fig. 2), these vectors are:

	Agl, Ain	Aps	Bpe, Bpu	Cbe	Fex	Lde, Pmu, Psy	Sau, Sfr	Tco	Fsy, Pab, Qpe, Qro	Ugl, Ula
<i>g3</i>	71 - 100	131 - 200	91 - 120	91 - 130	161 - 300	151 - 300	81 - 100	131 - 280	201 - 300	101 - 160
<i>g2</i>	21 - 70	71 - 130	51 - 90	61 - 90	81 - 160	61 - 150	51 - 80	71 - 130	101 - 200	51 - 100
<i>g1</i>	11 - 20	41 - 70	21 - 50	41 - 60	26 - 80	21 - 60	26 - 50	36 - 70	61 - 100	36 - 50
<i>v</i>	6 - 10	21 - 40	6 - 20	16 - 40	16 - 25	11 - 20	11 - 25	16 - 35	21 - 60	21 - 35
<i>im</i>	1 - 5	1 - 20	1 - 5	1 - 15	1 - 15	1 - 10	1 - 10	1 - 15	1 - 20	1 - 20
<i>g3</i>	61 - 80	121 - 160	81 - 110	91 - 110	136 - 205	61 - 80	116 - 145	126 - 165	221 - 300	101 - 130
<i>g2</i>	36 - 60	91 - 120	61 - 80	71 - 90	86 - 135	36 - 60	91 - 115	96 - 125	151 - 220	71 - 100
<i>g1</i>	16 - 35	71 - 90	41 - 60	51 - 70	56 - 85	16 - 35	71 - 90	76 - 95	101 - 150	51 - 70
<i>v</i>	11 - 15	41 - 70	21 - 40	26 - 50	36 - 55	11 - 15	36 - 70	46 - 75	51 - 100	41 - 50
<i>im</i>	1 - 10	1 - 40	1 - 20	1 - 25	1 - 35	1 - 10	1 - 35	1 - 45	1 - 50	1 - 40
<i>g3</i>	old, mature, young generative virginal im mature					<i>n (o)</i>	<i>n</i> ormal (<i>o</i> ppressed) conditions			
<i>g2</i>										
<i>g1</i>										
<i>v</i>										
<i>im</i>										
(c)	Aal	<i>Abies alba</i> (silver fir)				Pab	<i>Picea abies</i> (Norway spruce)			
	Agl	<i>Alnus glutinosa</i> (black alder)				Pmu	<i>Pinus mugo</i> (mountain pine)			
	Ain	<i>Alnus incana</i> (grey alder)				Psy	<i>Pinus sylvestris</i> (Scots pine)			
	Aps	<i>Acer pseudoplatanus</i> (sycamore)				Qpe	<i>Quercus petraea</i> (sessile oak, durmast)			
	Bpe	<i>Betula pendula</i> (silver birch)				Qro	<i>Quercus robur</i> (English oak)			
	Bpu	<i>Betula pubescens</i> (white birch)				Sau	<i>Sorbus aucuparia</i> (mountain ash, rowan)			
	Cbe	<i>Carpinus betulus</i> (hornbeam)				Sfr	<i>Salix fragilis</i> (crack willow)			
	Fex	<i>Fraxinus excelsior</i> (European ash)				Tco	<i>Tilia cordata</i> (small-leaved lime)			
	Fsy	<i>Fagus sylvatica</i> (beech)				Ugl	<i>Ulmus glabra</i> (wych (witch) elm)			
	Lde	<i>Larix decidua</i> (larch)				Ula	<i>Ulmus laevis</i> (white elm)			

Fig. 2 (Continued)

$$m_{\min} = [6 \ 12 \ 12 \ 12 \ 12 \ 7 \ 5 \ 12 \ 8 \ 4 \ 12 \ 5 \ 12 \ 2 \ 3 \ 2 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1],$$

$$m_{\max} = [7 \ 16 \ 16 \ 16 \ 16 \ 11 \ 13 \ 16 \ 10 \ 10 \ 16 \ 13 \ 16 \ 3 \ 5 \ 3 \ 3 \ 3 \ 2 \ 2 \ 2 \ 4 \ 2 \ 2].$$

To demonstrate how the components of vectors $\mathbf{m}$ are filled in, transitions $28 \rightarrow 22 \rightarrow 15 \rightarrow 9 \rightarrow 2$ could be considered (Fig. 2).

As estimated from the field observations (Logofet, 2001; Benabdellah's vegetation data), state 28 lasts from 5 to 20 years:

$$m_{\min} (28-6) = m_{\min} (22) = 1, \quad m_{\max} (28-6) = m_{\max} (22) = 4.$$

Since the climax states last until a drastic disturbance, these first six elements are not presented in the vectors. To estimate duration of the herb stage, geobotanic data have been used, while the durations of ontogenetic

intervals have been applied for the subsequent stages with a developed tree layer. Minimal and maximal duration times for the next state 22 are:

$$(v-im|\mathbf{Sau}, n) = (11-1) \text{ and } (v-im|\mathbf{Sau}, n) = (25-10) \text{ years: } 10 < (v-im|\mathbf{Sau}, n) < 15.$$

Therefore, state 22 lasts from 10 to 15 years or (in units of Δt) from 2 to 3 steps:

$$m_{\min} (22-6) = m_{\min} (16) = 2, \quad m_{\max} (22-6) = m_{\max} (16) = 3.$$

Virginal ontogenetic age v is used in the formulae for pioneer species (rowan, birches, hornbeam, willow) and young generative age $g1$ is set for the others.

States 15 and 9 have the following duration times:

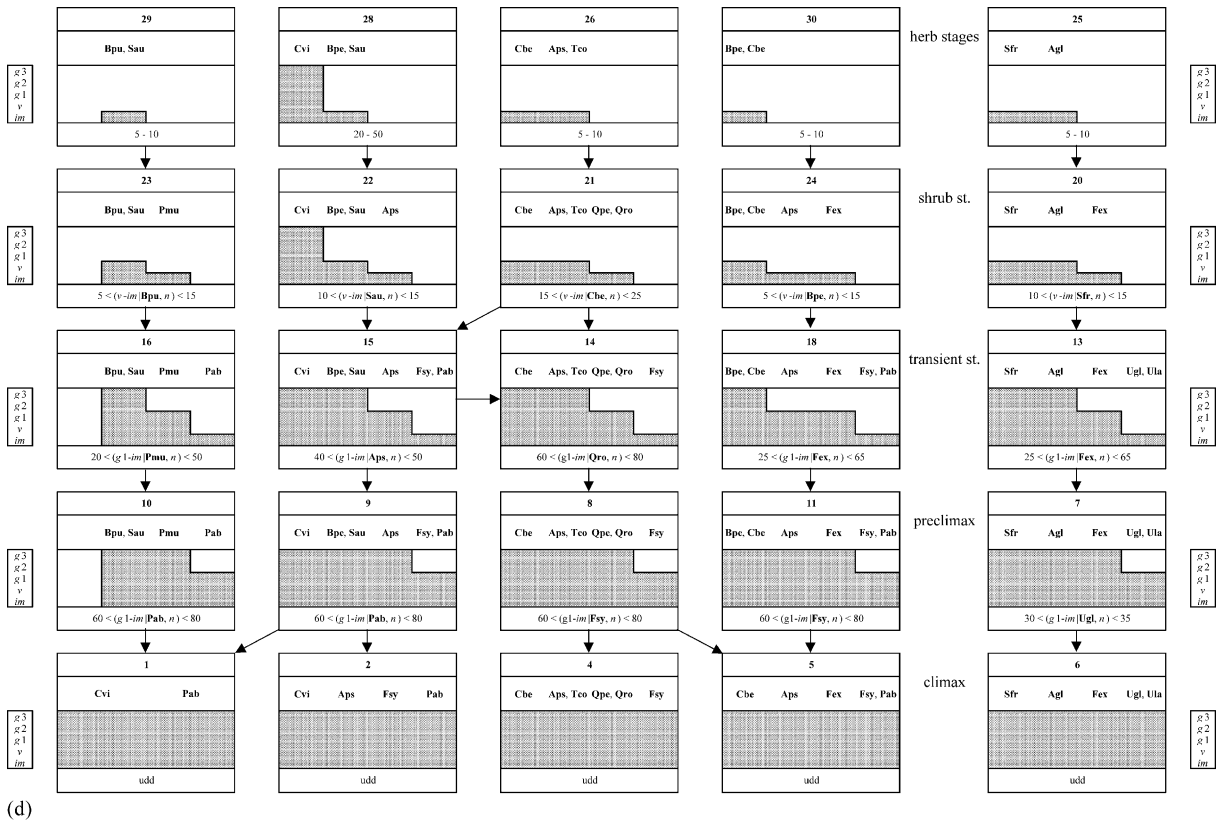


Fig. 2 (Continued)

$(g1-im|Aps, n) = (41-1)$ and $(g1-im|Aps, n) = (70-20)$ years: $40 < (g1-im|Aps, n) < 50$, $m_{\min} (15-6) = m_{\min} (9) = 8$, $m_{\max} (15-6) = m_{\max} (9) = 10$ and $(g1-im|Pab, n) = (61-1)$ and $(g1-im|Pab, n) = (100-20)$ years: $60 < (g1-im|Pab, n) < 80$, $m_{\min} (9-6) = m_{\min} (3) = 12$, $m_{\max} (9-6) = m_{\max} (3) = 16$.

Climax state 2 lasts until a drastic disturbance.

Obviously, p_{ii} stands for the probability to remain in state i . If there is only one possibility for leaving this state, then corresponding probability equals to $(1-p_{ii})$. When there are more ways to leave state i (a ramification point), the value of $(1-p_{ii})$ has to be distributed among those possible transitions. The chosen distribution va-

lues can be presented in the branch (row-stochastic) matrix B .

In the case shown in Fig. 2, only the sum of off-diagonal elements in the 8th, 9th, 15th, 21th and 24th rows should be split-up by means of additional geobotanic assumptions:

$$b_{84} = 0.99, b_{85} = 0.01; b_{91} = b_{92} = 1/8, b_{93} = 3/4; b_{15,9} = b_{15,14} = 1/2; b_{21,14} = b_{21,15} = 1/2; b_{24,17} = b_{24,18} = b_{24,19} = 1/3.$$

Then, the off-diagonal elements in the transition probability matrices $P_{\min}$ and $P_{\max}$ can be calculated as:

$$p_{ij,\min} = (1-p_{ii,\min}) \cdot b_{ij}, p_{ij,\max} = (1-p_{ii,\max}) \cdot b_{ij}, i, j = 1, \dots, 30, i \neq j \quad (2)$$

The initial area-distribution vector x is selected in our case (Fig. 2) in a way that the final (climax)

area-distribution vector y_{mod} is close to y_{PNV} (potential natural vegetation) distribution. PNV distribution was obtained from another project group (P.A. Schmidt, Personal communication). Thus, these three vectors are:

$x = [0 \dots 0 \ 0.003 \ 0.285 \ 0.356 \ 0.356 \ 0 \ 0]$ (the first 24 elements equal to zero),

$y_{\text{mod}} = [0.053 \ 0.053 \ 0.321 \ 0.564 \ 0.006 \ 0.003 \ 0 \dots 0]$ (the last 24 elements equal to zero),

$y_{\text{PNV}} = [0.030 \ 0.052 \ 0.357 \ 0.553 \ 0.005 \ 0.003 \ 0 \dots 0]$ (the last 24 elements equal to zero).

Vegetation dynamics of the modelled succession process for some selected states are shown in Fig. 3. It demonstrates that the pioneer stages (28, 22), after a maximum extension in the beginning, occupy less and less area, the intermediate states grow up to a maximum after about 100 years and then their relative area also decreases continuously, while the area occupied by the climax communities (4) is monotone increasing. It is

because the assumed minimal duration time for the herb stage is only 25% of its maximal duration time that a marked difference appears between the curves for *Calamagrostis villosa* on the left and on the right of Fig. 3. The Figure demonstrates also that the greatest changes during this special succession process occur during the first 300 years.

4. Inhomogeneous model for forest successions in the areas polluted by SO_2

Strong SO_2 pollution happened in the last two decades in Eastern Erzgebirge area. The concentration reached its maximum at the end of the 1980s (Abraham et al., 2000). The spruce stands, especially on high altitude, returned to the herb and grass stages because of pollution damage followed by clear felling.

Successions after damage by SO_2 pollution have been modelled for two areas, which are located

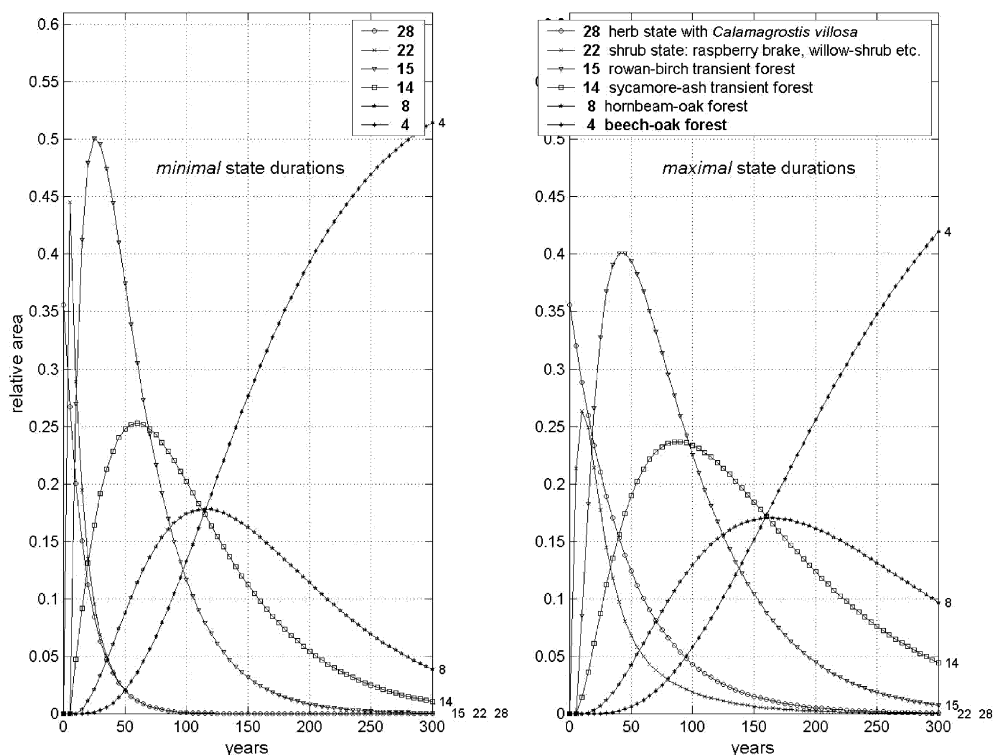


Fig. 3. Time dependence of relative areas for typical vegetation development in the Erzgebirge (after the scheme of Fig. 2).

near Altenberg (latitude 50°44'N, longitude 13°45'W) and Olbernhau (latitude 50°39'N, longitude 13°20'W), in forest districts at the border to the Czech Republic, at the altitude between 780 and 890 m. Both areas are very similar in geobotanical conditions. Therefore, the same model could be applied for them. Before SO₂ pollution in the last two decades, there was a spruce stand on the basic rock of quartz-porphry and gneiss. Secondary successional changes after clear felling of devastated forests in both areas are shown in Figs. 4 and 5.

The following plant communities of state 1 (Figs. 4 and 5) are the first to settle on clear-cut areas (Schubert et al., 1995): *Epilobio angustifolii*–*Digitalietum pupureae* (often on acid and poor nutrient soils in submontane and montane regions); *Treintali europaea*–*Calamagrostium villosae* (on acid soils of frequent or heavy precipitation with dominance of *C. villosa* and *T. europea*); *Myrtillo*–*Avenelletum flexuosae* (on acid soil with dominance of *A. flexuosa* (*Dechampsia flexuosa*) and *Vaccinium myrtillus*); *Epilobio angustifolii*–*Senecionetum fuchsii* (small sites of lime melioration with rich to neutral soil and dominance by *Senecio fuchsii*). As estimated in the field, state 6 can last from 5 to 15 years in our case.

Succession has now three possibilities (nos. 2, 3, 4) to go further (Figs. 4 and 5). Realisation of a particular trajectory depends on the answer to a question at what degree the juveniles of pioneer trees (birches, rowan) can survive under the canopy of reedgrass (*C. villosa*), which predominates and suppresses all other species on open areas. If the reedgrass wins the competition, then transition to the grass stage with dominance of *C. villosa* happens. According to the field observations, this state lasts more than 20 years and the succession process has not yet coped to overcome it. As geobotanic descriptions show (Logofet, 2001; Benabdellah's vegetation data), *C. villosa* and *A. flexuosa* codominate in the herb layer of study areas.

Durations of the other states are calculated by the method mentioned above, according to the species-specific ontogenetic development in the area (Fig. 5):

$$\begin{aligned} \mathbf{m}_{\min} &= [5 \ 20 \ 5 \ 25 \ 60 \ \text{udd}], \ \mathbf{m}_{\max} \\ &= [15 \ 25 \ 15 \ 40 \ 80 \ \text{udd}] \end{aligned}$$

In the time-inhomogeneous version of the model, the transition probabilities vary at every time step in a way such that each state n ($n = 3, \dots, 6$) would continue for some random time between $\mathbf{m}_{\min}(n)$ and $\mathbf{m}_{\max}(n)$ years.

Initial area-distribution vector $\mathbf{x}$ was obtained from Benabdellah's vegetation data (Logofet, 2001):

$$\mathbf{x} = [0 \ 1 \ 0 \ 0 \ 0 \ 0]$$

At least three major alternative scenarios are proposed, namely, with hard, soft, and without any, anthropogenic impact (Fig. 5). The difference between scenarios is expressed by nontrivial elements of the corresponding branch matrices $\mathbf{B}_{\text{an}}$, $\mathbf{B}_{\text{hf}}$ and $\mathbf{B}_{\text{na}}$:

$$\begin{aligned} b_{53} &= p, \ b_{54} = 1 - p \ (0 \leq p \leq 1); \ b_{63} = q, \ b_{65} \\ &= 1 - q \ (0 \leq q \leq 1) \end{aligned}$$

More rigorously, if p and q indicate the relative areas of states 5 and 6, which are under alternative management, then the anthropogenic impact is estimated as:

$$\begin{aligned} \text{hard} \quad \text{when } p = q = 1: \quad &b_{\text{an},53} = 1, \ b_{\text{an},54} = 0, \\ &b_{\text{an},63} = 1, \ b_{\text{an},65} = 0 \end{aligned}$$

(ploughing, liming and seed raining imply that transitions $6 \rightarrow 3$ and $5 \rightarrow 3$ occur in one step);

$$\begin{aligned} \text{soft} \quad \text{when } p = q = 1/2: \quad &b_{\text{hf},53} = 1/2, \ b_{\text{hf},54} = 1/2, \\ &b_{\text{hf},63} = 1/2, \ b_{\text{hf},65} = 1/2 \text{ and} \\ \text{absent} \quad \text{when } p = q = 0: \quad &b_{\text{na},53} = 0, \ b_{\text{na},54} = 1, \\ &b_{\text{na},63} = 0, \ b_{\text{na},65} = 1. \end{aligned}$$

The other nontrivial elements of these matrices remain the same: $b_{15} = b_{16} = 1/2$.

Transition probability matrices $\mathbf{P}$ at every chain step are thereafter calculated by Eqs. (1) and (2).

Time dynamics of relative areas for all vegetation states (Altenberg, Olbernhau) are shown in Figs. 6 and 7.

For the Altenberg area a second model version was constructed to estimate the fact of management activities (Fig. 7). The corresponding scheme is given in Fig. 5. The main difference in compar-

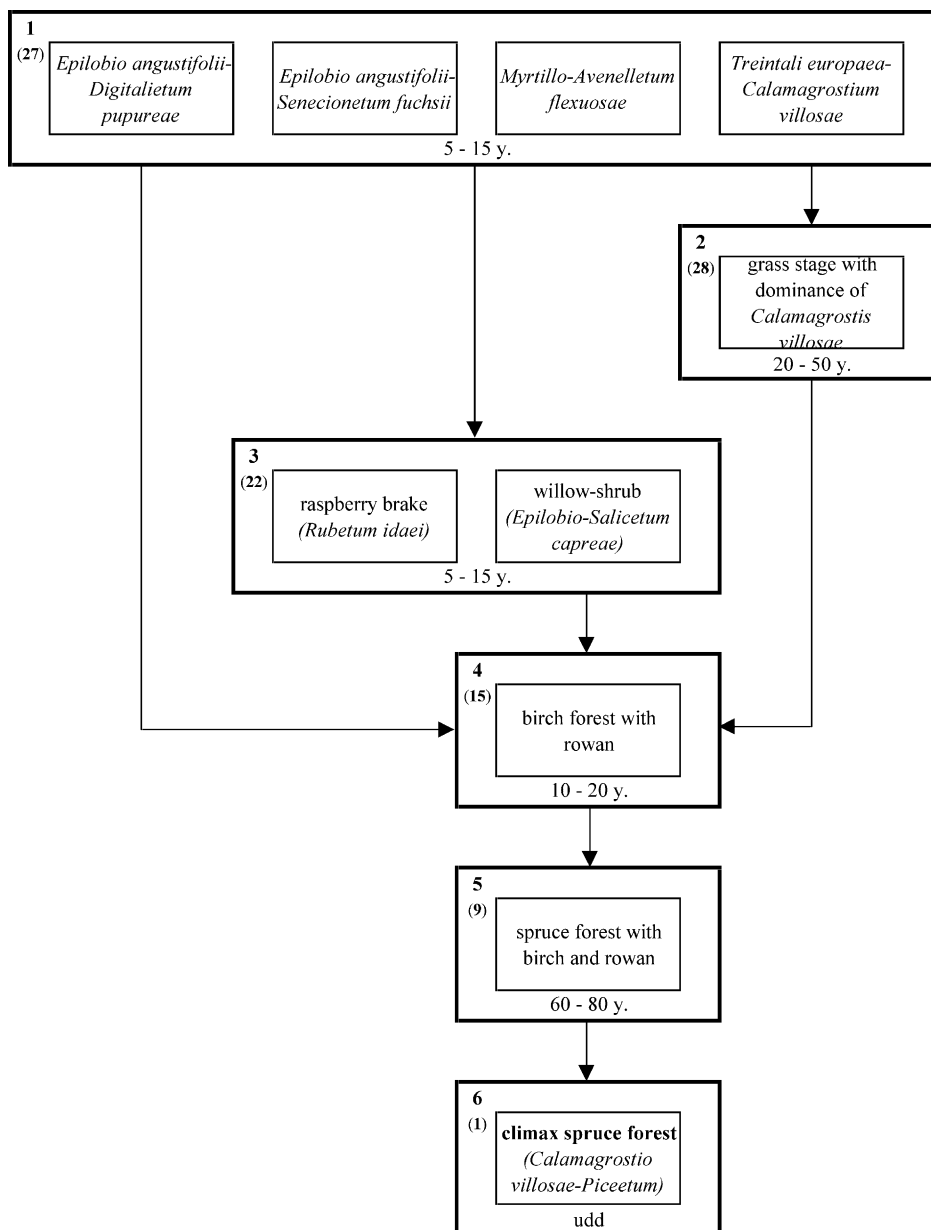


Fig. 4. Possible secondary successional changes after SO₂ pollution of spruce stands in the areas near Altenberg and Olbernhau. Parenthetical numbers correspond to state enumeration of Fig. 2, state-duration times are given at the bottom in years (state 6 lasts until a drastic disturbance). See Fig. 2b for explanation of 'udd'.

ison with Fig. 4 is the appearance of an additional unstable stage 7. This is a consequence of management activities, such as ploughing and seeding, to shorten the duration of the grass stage 2 (*C. villosa*). As a consequence, an unstable even-aged

spruce forest (stage 7) appears. If there is no strong disturbance, this forest transfers in time to the climax spruce forest (stage 6). It is also possible that stage 7 goes back to stage 1 after disturbance (for instance, strong snow events). A transition

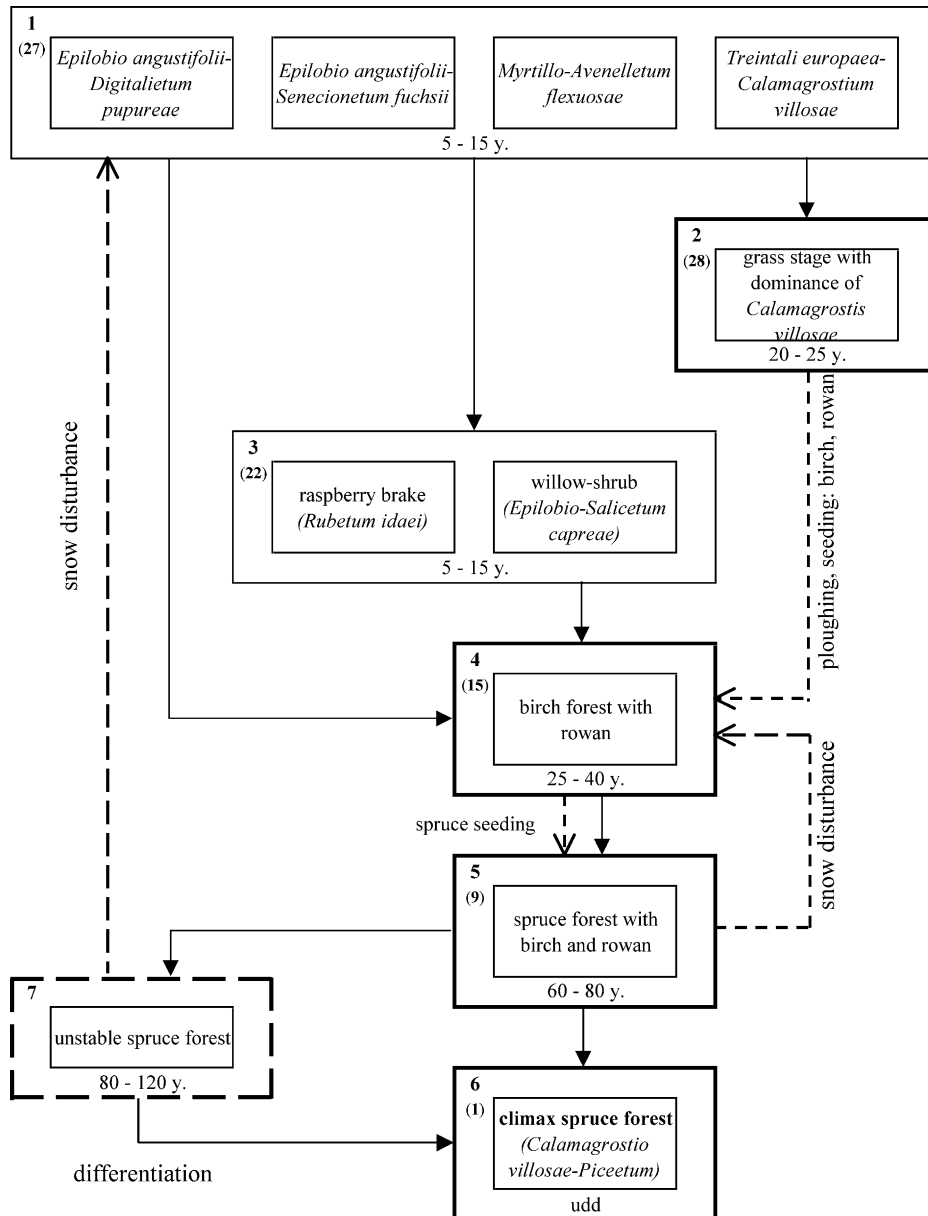


Fig. 5. Possible successional changes under human impacts (ploughing, seeding) and natural disturbances (snow catastrophes) after SO_2 pollution of spruce stands in the areas near Altenberg and Olbernhau. Dashed arrows show anthropogenic influence, solid ones denote development without human impact; parenthetical numbers correspond to state enumeration of Fig. 2, state-duration times are given at the bottom in years (states 2 and 6 last until a drastic disturbance).). See Fig. 2b for explanation of 'udd'.

from stage 5 to 4 in this model version is considered to take into account possible snow events as well. The spruce seeding is introduced to accelerate the transition from stage 4 to 5.

The initial-area distribution vector has been taken again from the collected field data. The results are presented in Fig. 7. The comparison between Figs. 7 and 6 shows that the stable climax

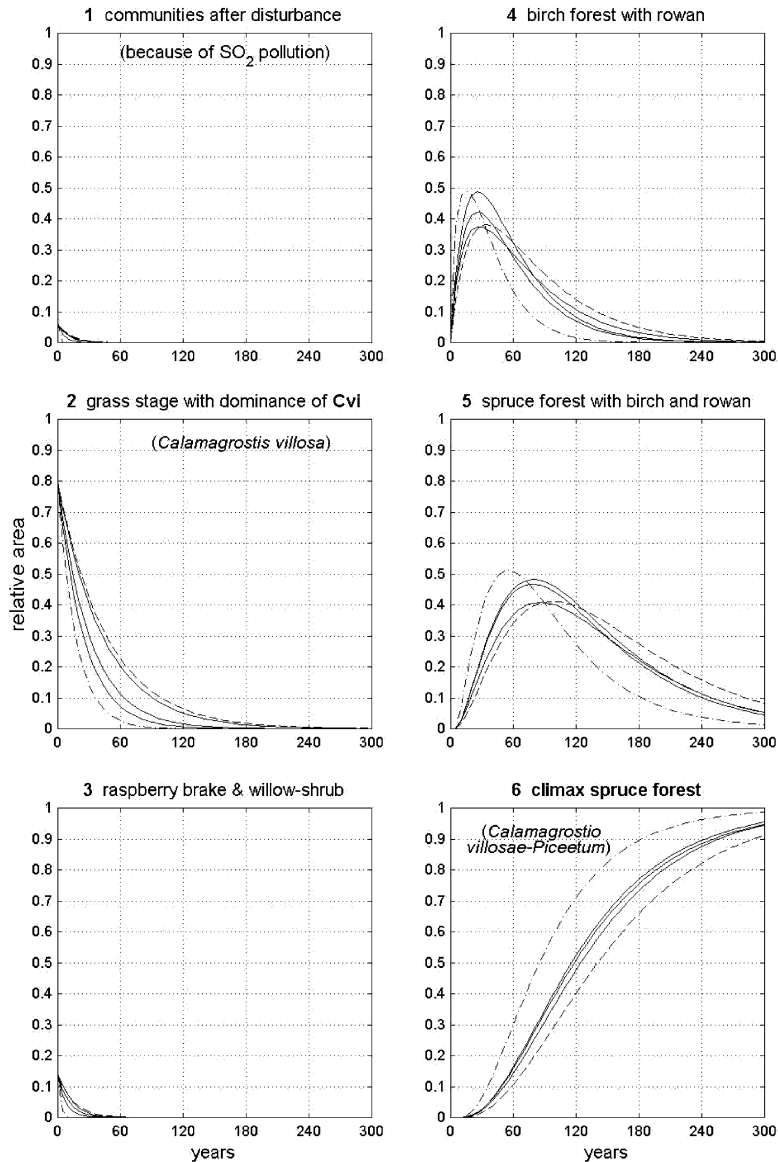


Fig. 6. Time dependence of relative areas for secondary succession stages in the area near Altenberg with stage durations as follows (see Fig. 4): minimal (dashdot), maximal (dashed) and randomly intermediate (solid lines).

spruce forest is obviously reached later in the case of management activities. The reason is the existence of an additional unstable spruce forest (stage 7). So, the sum of stages 6 and 7 in Fig. 7 corresponds to stage 6 in Fig. 6 in the case of management.

An important fact for management strategies can be observed by further comparison between Figs. 6 and 7 (stage 2). It shows that ploughing considerably helps overcome the grass stage and realise an earlier start of tree stages in this way.

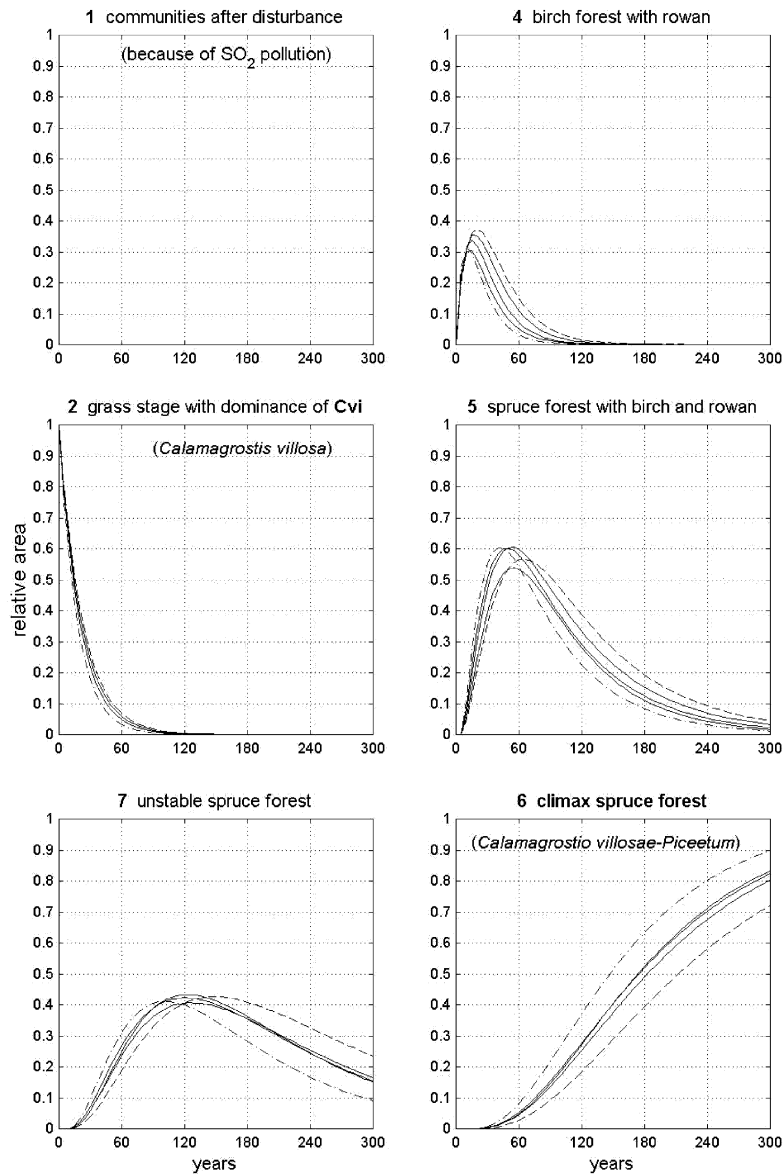


Fig. 7. Time dependence of relative areas under specific forest management in the area near Altenberg with stage durations as follows (see Fig. 5): minimal (dashdot), maximal (dashed) and randomly intermediate (solid lines).

5. Discussion

The succession process implies selection of serial plant communities which enable, by successively substituting one for another (Fig. 2), formation or restoration (Figs. 4 and 5) of the forest community and its proper environment after disturbances. Other components of the forest ecosystem, such

as soil, although involved in succession, do not necessarily have the same, nor synchronised, time characteristics (Targul'yan and Sokolov, 1978). Model application enables estimating the time it would take for a complete forest mosaic to develop over the Erzgebirge area. This time varies from 120 (Fig. 3, left) to 150 years (Fig. 3, right). If the goal is to restore the vegetation cover in the Erzgebirge,

the Figures reveal that a territory is needed for the serial forest communities—the birch and maple ones—to develop during the nearest 50 years. They make the ecotopes ready for inhabiting by, and development of, the climax species: oaks (*Quercus robur* and *Q. petraea*) and beech (*Fagus sylvatica*). However, the birch and maple forests are of low forestry value, and, if they are excluded by the spruce forest culture, then one should not expect any climax forest to develop since the vegetation of the area would now lose its inherent potential to renew the broad-leaved forest.

Neglecting the natural course of forest succession has now brought about formation of a great number of subclimax communities as a result of intensive forestry in the area. The former strategy of forest management would result in complete destruction of long-derivative stages of the beech and hornbeam-oak forest successions, as it has already happened with the montane fir–spruce forests.

It is worth noting that a positive experience does exist in restoration of beech and oak forests. In the long-term experiments laid down by Dokuchaev (1898) and observations by Morozov (1925), Kaltutskii et al. (1972) and Arbuzova (1995), it is shown that initiation (for 40 years) and restoration of broad-leaved forest are quite possible.

Future states of a forest can be predicted from the general knowledge of succession regularities applied to particular forest conditions. Under current anthropogenic changes in the vegetation, the method of analogy is normally applied, which enables constructing series of secondary succession (Ludi, 1958; Petrov, 1974). However, the life span of a researcher/observer is limited and principally less than the time scale of forest successions. Numeric experiments with models are therefore crucial for prediction of tendencies in forest development. For example, a phenomenon such as toxicogenic impact on forest was revealed about 20 years ago. Model studies of the effects this kind of stress induces in the coniferous forest area of the Erzgebirge have shown that six stages can be specified in the forest recovery succession (Fig. 4). A fairly narrow set of initial groupings of species remained after the tree layer had been

removed from the devastated forest; those groupings are followed either by a meadow stage (typically a reedgrass or tufted-hair-grass meadow), which can still be observed in some spots, or by a shrub stage (severely reduced), or by a birch forest stage. At present, 25 years after the secondary succession started, a bilberry birch forest with young growth of spruce has formed near Olbernhau, while the succession still remains at the meadow stage near Altenberg. In the latter area, an attempt was made to initiate a spruce forest stage by plantation of spruce and rowan-tree seedlings, but it will function as a meadow community at least for 20 years more, until the spruce canopy will close. According to the scheme of Fig. 4, the stage of *C. villosae*–*Piceetum*, which is close to the climax one, *Vaccinio*–*Piceetum*, can hardly be expected within the nearest 100 years.

A numeric experiment for this area of coniferous forest in the Erzgebirge demonstrates that:

- 1) duration of the reedgrass meadow stage increases (Fig. 6.2);
- 2) the shrub stage reduces (Fig. 6.3);
- 3) the birch forest stage reaches its maximum in 25 years (Fig. 6.4);
- 4) the mixed birch-spruce forest stage reaches its maximum in 70–75 years (Fig. 6.5);
- 5) the reedgrass spruce forest forms in 110 years with a 50% probability.

All these points conform to the geobotany views of a territory under multiple anthropogenic impacts (Tishkov, 1979; Razumovsky, 1981; Armand, 1992): if the vegetation has several times passed through an incomplete cycle of regeneration succession then its initial stages become longer; reduction of the shrub stage is also a result of spruce monoculture forest management.

Disturbances of the tree stand in a montane spruce forest, when caused by snow, threw the succession back to its initial stage or to the birch forest stage (Fig. 5). Planting birch and rowan trees initiates formation of a seral birch forest. The situation is mirrored by a model experiment shown in Fig. 7. The probability for a subclimax-type, spruce forest community (*C. villosae*–*Piceetum*) to develop will not exceed 0.2 even at the end of the

21st century (Fig. 7.6), while the mixed birch-spruce forest remains dominant in this area of coniferous forest during the whole century (Fig. 6.5). It means that the retardation tendency in the forest recovery does continue. In other words, an impact on the course of succession may induce centennial delay, i.e. a long persistence of vegetation in the initial stages of succession.

6. Conclusion

The preliminary application of Markov chain models in the Erzgebirge has shown that such models may be helpful for quantified estimations of the long-term development of forest areas and so they could be relevant for forest management decisions. Because nearly all forests are man-made in the Erzgebirge, seldom can one find out an area under succession. This makes a comparison of model results with real vegetation distribution quite difficult or impossible. The few geobotanical data collected in this area have been mainly used for determination of the initial conditions for the model runs. The aim of this paper is to show that Markov chain models could be used in principle for estimations and management decisions in the Erzgebirge as well. For a successful use of this possibility the models have to be validated and calibrated by experts. Up to now only qualitative tests have been performed, e.g. it was tested that the results of the first model do not contradict with PNV results (potential natural vegetation, geobotanic estimations obtained independently by another project group: Schmidt et al., 2001). It follows from the preliminary results obtained with the 2nd and 3rd models that the tendencies and features revealed include increase in the times needed for both coniferous and broad-leaved forests to recover after drastic disturbances (via increments in duration of the initial stages of succession), expansion of the area for serial vegetation, and formation of a spruce subclimax which is distinct from the natural one. These findings demonstrate a need to change the strategic aims in forest management: in the 21st century, the nature-exploitation policy should include the idea of montane forest restoration in the Erzgebirge.

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